

Curriculum Vitae – Xuancheng Ye

CONTACT INFORMATION	Courant Institute of Mathematical Sciences New York University	xy2589@nyu.edu +1 (929) 655-8731
EDUCATION	Courant Institute of Mathematical Sciences, NYU M.S. in Mathematics ● GPA: 3.9/4.0 School of Mathematics, Southwestern University of Finance and Economics (SWUFE) <i>Guanghua Innovative Talents Experimental Class</i> B.S. in Math & Applied Maths, B.Ec. ● GPA: 87.77/100 ● Ranking: 3/48 (SWUFE Outstanding Graduate, SWUFE Merit Student)	
RESEARCH INTERESTS	Theory and Methods: Optimal Transport; Applied Probability; Optimization; Causal inference Applications: Intervention optimization; the multiphase optimization strategy (MOST)	
ACADEMIC PAPERS	Published/Submitted [1] Andrew D. Lipnick*, Esteban G. Tabak*, Giulio Trigila*, Yating Wang*, <u>Xuancheng Ye*</u> , and Wenjun Zhao*. (2025). "The Monge optimal transport barycenter problem" (arXiv preprint arXiv:2507.03669) Under minor revision at <i>International Statistical Review</i> . [2] Chang Liu, Bowen Deng, Liang Wu, <u>Xuancheng Ye</u> , and Min Guo. (2026). "Multi-Agent Reinforcement Learning Dynamics in Event-Driven Limit Order Environments" Submitted to <i>Applied Soft Computing Journal</i> . [3] Siyu Heng*, <u>Xuancheng Ye*</u> , Jillian Strayhorn, Charles M. Cleland, Lalitha Parmeswaran, and Marya Gwadz. (2025). "Selecting optimized behavioral interventions from an optimization randomized controlled trial on increasing COVID-19 testing for African American/Black and Latino frontline essential workers not up-to-date on COVID-19 vaccination" <i>Annals of Behavioral Medicine</i> . In press (selected as <i>Editor's Choice</i>) [4] Chang Liu, Bowen Deng, <u>Xuancheng Ye</u> , and Min Guo. (2025). "Intelligent Decision Making under Conflicting and Cooperative Scenarios with incomplete information: a wresting of Banks' Loan Policies and Firms' repayment behaviors." <i>Applied Soft Computing Journal</i> .173(2):112898 In preparation [5] Esteban G. Tabak*, <u>Xuancheng Ye*</u> , and Wenjun Zhao*. (2025). "Time series analysis through the optimal transport barycenter problem" [6] Marya Gwadz, Campos, S., Prema Filippone, Robinson, J., Chero, R., Siyu Heng, Lalitha Parmeswaran and <u>Xuancheng Ye</u> . (2025). "Intervention components to increase COVID-19 testing for African American/Black and Latine frontline essential workers not up-to-date on vaccination: Exploring acceptability and feasibility"	
RESEARCH EXPERIENCE	Graduate Researcher Courant Institute of Mathematical Sciences, New York University 08/25 - now "Optimal Transport Component Analysis through Tensorial PCA" Advisor: Prof. Esteban G. Tabak (see Academic Paper [4]) ● Proposed an adversarial formulation to solve the Monge optimal transport barycenter problem	

- Used the tensorial factorization to represent the independence between the transformed outcome y and factor z : using an adaptive, factorized construction for $F(z)$ and $G(y)$

Graduate Researcher | Courant Institute of Mathematical Sciences, New York University **01/24 - 07/25**
 Advisor: [Prof. Esteban G. Tabak](#) | (see Academic Paper [1])

- Proposed a general methodology to solve the data-driven Monge optimal transport barycenter problem
- Relaxed independence to the uncorrelation between all functions of factor z and the transformed outcome y within suitable finite-dimensional spaces

Graduate Researcher | Courant Institute of Mathematical Sciences, New York University **05/24 - 09/24**
 "Variational Inference Frameworks for Distribution Estimations and Inverse Problem Formulation"
 Advisor: [Prof. Georg Stadler](#)

- Analytically derived the VI model for linear Gaussian high-dimensional regression problems.
- Developed a JAX-based Variational Inference framework for ELBO gradient computation. Enabled the use of a mixture of Gaussians for estimation.

Research Assistant | Department of Biostatistics, New York University **07/24 - 06/25**
 Advisor: [Prof. Siyu Heng](#) & [Prof. Marya Gwadz](#) | (see Academic Paper [2], [5])

- Applied the multi-phase optimization strategy (MOST) to a 2^4 factorial optimization trial.
- Proposed an alternative posterior expected value approach and used MC simulation for evaluation.
- Proposed three missingness imputation strategy for composite outcome in MOST.

Undergraduate Researcher | Southwestern University of Finance and Economics **09/22 - 09/23**
 Advisor: [Prof. Chang Liu](#) | (see Academic Paper [3])

- Provided a novel approach to simulating dynamic decision-making processes in the context of interactions between banks and firms under conditions of information asymmetry and competition.
- Introduced an intelligent game arena designed to model and analyze the decision making processes of banks and firms operating with incomplete information.
- Determined optimal strategies and actions by linking actions and resultant rewards through Hybrid Archimedean Copula (HAC) and Multi-objective Integer Quadratic Programming (MIQP).

WORKSHOP **Computational Summer Workshop** | Lawrence Livermore National Laboratory **05/25 - 06/25**
 Advisor: Prof. Monique Chyba.

- Two weeks intensive workshop on numerical methods and high-performance computing (HPC).

AWARDS Leader of China National College Students' Innovation Training Project (\$1500) 2021
 Research Innovation Scholarship, SWUFE 2021,2023
 Academic Excellence Scholarship, SWUFE 2019, 2020, 2021, 2022, 2023
 Innovation and Entrepreneurship Scholarship, SWUFE 2020,2022

TEACHING EXPERIENCE **Grader of CSCI-UA 421 Numerical Computing** Spring 2025
Courant Institute of Mathematical Sciences, NYU Instructor: [Prof. Michael L. Overton](#)
CA of GPH-GU 2363 Causal Inference: Design and Analysis(Graduate-level) Fall 2024, Fall 2025
Department of Biostatistics, NYU Instructor: [Prof. Siyu Heng](#)

SERVICE Reviewer of **Journal of Biopharmaceutical Statistics** (2), **Agriculture & Food Security** (2)

SKILLS Python, Matlab, R, Julia